



Digital Egypt Pioneers Initiative (DEPI)

Traffic Accident Analytics Dashboard Using Power BI

Data Analytics Project

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Tools Used:
• Microsoft Power BI

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Abstract

Traffic accidents are among the leading causes of injuries and economic losses worldwide, making data analysis an essential tool for understanding accident patterns and supporting better decision-making.

This project presents an interactive **Traffic Accident Analytics Dashboard** developed using **Microsoft Power BI**. The dashboard analyzes traffic accident data from multiple perspectives, including weather conditions, road types, road conditions, driver characteristics, vehicle types, traffic density, speed limits, and accident severity.

The project involved data cleaning and transformation using **Power Query**, followed by the creation of dynamic **DAX measures** to calculate key performance indicators (KPIs). Interactive visualizations and slicers were implemented to allow users to explore the data efficiently and gain meaningful insights.

The developed dashboard enables decision-makers to identify accident trends, compare different risk factors, and better understand the relationships between environmental conditions, driver behavior, and accident occurrence. The project demonstrates how Business Intelligence tools can transform raw data into valuable insights that support data-driven decision-making.

Keywords: Power BI, Traffic Accidents, Data Analytics, Dashboard, Power Query, DAX.

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Chapter 1

[Introduction]

1.1 Project Background

Road traffic accidents are among the leading causes of injuries, fatalities, and economic losses worldwide. Understanding the factors that contribute to traffic accidents is essential for improving road safety and supporting data-driven decision-making.

Business Intelligence (BI) tools such as **Microsoft Power BI** enable analysts to transform raw accident data into meaningful insights through interactive dashboards, allowing organizations and decision-makers to monitor accident patterns and identify high-risk conditions.

This project develops an interactive Power BI dashboard to analyze traffic accident data and highlight the impact of weather conditions, road characteristics, driver behavior, vehicle types, and environmental factors on accident occurrence and severity.

1.2 Problem Statement

Traffic accident data often contains valuable information, but without proper analysis it becomes difficult to identify the main causes of accidents or recognize hidden patterns.

Traditional reports provide only static information, making it difficult to compare accident trends across different conditions such as weather, road type, or driver characteristics.

Therefore, there is a need for an interactive analytical dashboard that enables users to explore accident data efficiently and supports better decision-making regarding road safety.

1.3 Project Objectives

The main objectives of this project are:

- I. Develop an interactive dashboard using Microsoft Power BI.
- II. Analyze traffic accident patterns.
- III. Study the impact of weather conditions on accident occurrence.
- IV. Analyze accident distribution based on road type.
- V. Analyze driver characteristics such as age, experience, and alcohol consumption.
- VI. Evaluate the effect of road conditions and traffic density.
- VII. Identify accident severity levels.
- VIII. Support data-driven decision-making through interactive visualizations.

1.4 Project Scope

This project analyzes historical traffic accident data using Power BI.

The analysis focuses on the following variables:

- Weather
- Road Type
- Road Condition
- Time of Day
- Road Light Condition
- Driver Age
- Driver Experience
- Driver Alcohol Consumption
- Vehicle Type
- Traffic Density
- Speed Limit
- Accident Severity
- Accident Occurrence

The project does not include predictive machine learning models or real-time streaming data. It focuses on descriptive and exploratory analytics using Power BI.

1.5 Project Tools

<u>Tool</u>	<u>Purpose</u>
Microsoft Power BI Desktop	Dashboard Development
DAX	Calculated Measures and KPIs
CSV Dataset	Data Source
Power Query	Data Cleaning and Transformation

1.6 Expected Outcomes

After completing this project, the dashboard enables users to:

- I. Monitor overall accident statistics.
- II. Compare accident frequencies across different weather conditions.
- III. Analyze accident severity.
- IV. Study driver behavior and demographics.
- V. Evaluate road and vehicle factors.
- VI. Identify accident trends using interactive reports.

1.7 Project Structure

The project consists of four interactive dashboards:

Dashboard 1 – Executive Overview

Provides an overall summary of accident statistics, weather conditions, road types, and accident severity.

Dashboard 2 – Driver Analysis

Focuses on driver age groups, driving experience, and alcohol-related accidents.

Dashboard 3 – Road and Vehicle Analysis

Analyzes road conditions, vehicle types, and traffic density.

Dashboard 4 – Environment and Time Analysis

Examines accidents based on time of day, lighting conditions, and speed limits.

Chapter 2

[Dataset Description]

2.1 Dataset Overview

This project is based on a **Traffic Accident Prediction Dataset**, which contains traffic accident records collected under different environmental, road, vehicle, and driver conditions. The dataset is used to analyze accident patterns and identify the factors that influence accident occurrence and severity.

The dataset contains **840 records** and **14 attributes**, representing various characteristics related to traffic accidents.

The data was imported into **Microsoft Power BI Desktop** in CSV format and prepared using **Power Query** before performing the analytical process.

2.2 Dataset Source

The dataset was obtained from **Kaggle** and was provided in CSV format for educational and analytical purposes.

Dataset Format:

- I. CSV (Comma-Separated Values)
- II. Analysis Tool:
- III. Microsoft Power BI Desktop
- IV. Data Preparation Tool:
- V. Power Query

2.3 Dataset Structure

Attribute	Data Type	Description
Weather	Text	Weather condition during the accident
Road_Type	Text	Type of road where the accident occurred
Time_of_Day	Text	Time period of the accident
Traffic_Density	Whole Number	Level of traffic density
Speed_Limit	Whole Number	Maximum speed limit of the road
Number_of_Vehicles	Whole Number	Number of vehicles involved
Driver_Alcohol	Whole Number	Indicates whether the driver consumed alcohol (0 = No, 1 = Yes)
Accident_Severity	Text	Severity level of the accident
Road_Condition	Text	Condition of the road surface
Vehicle_Type	Text	Type of vehicle involved
Driver_Age	Whole Number	Driver's age
Driver_Experience	Whole Number	Years of driving experience
Road_Light_Condition	Text	Road lighting condition
Accident	Whole Number	Accident occurrence (Target Variable)

2.4 Data Preparation Overview

Before creating the dashboard, the dataset was examined and prepared to ensure data quality and consistency.

The preparation process included:

- I. Verifying column names.
- II. Assigning appropriate data types.
- III. Removing blank rows.
- IV. Filtering unnecessary records.
- V. Creating new analytical columns (Age Group and Experience Group).
- VI. Preparing the dataset for visualization and DAX calculations.

2.5 Created Columns

To improve the analysis, two additional columns were created in Power Query.

Age Group

Drivers were classified into age categories to simplify comparison.

Driver Age	Age Group
Less than 26	Youth
26–35	Adult
36 and above	Old

Experience Group

Driver experience was categorized into groups.

Experience	Group
0–5 Years	Beginner
6–10 Years	Intermediate
More than 10 Years	Experienced

2.6 Dataset Purpose

The dataset was analyzed to answer several business questions, including:

- I. Which weather condition records the highest number of accidents?
- II. Which road type experiences the highest accident frequency?
- III. How does driver age affect accident occurrence?
- IV. Does alcohol consumption influence accident frequency?
- V. Which vehicle type is involved in the largest number of accidents?
- VI. How do road conditions and traffic density affect accidents?
- VII. What is the distribution of accident severity levels?

Chapter 3

Data Cleaning and Transformation

3.1 Data Preparation Process

Before building the dashboard, the dataset was cleaned and transformed using **Power Query** in Microsoft Power BI. Data preparation is considered one of the most important stages in any Business Intelligence project because it improves data quality and ensures accurate analytical results.

The following steps were performed before loading the data into the Power BI model.

3.2 Importing the Dataset

The traffic accident dataset was imported into Microsoft Power BI Desktop in **CSV format**.

After importing the data, Power Query was used to inspect and transform the dataset before loading it into the data model.

3.3 Data Type Validation

Each column was assigned the appropriate data type to ensure that Power BI could perform calculations and visualizations correctly.

Examples include:

Column	Data Type
Weather	Text
Road Type	Text
Driver Age	Whole Number
Driver Experience	Whole Number
Speed Limit	Whole Number
Traffic Density	Whole Number
Accident	Whole Number

Assigning the correct data types is essential because incorrect data types may lead to calculation errors and inaccurate visualizations.

3.4 Removing Blank Rows

Blank rows were identified and removed from the dataset to eliminate incomplete records that could affect the analysis.

Removing blank records improves data consistency and prevents inaccurate calculations during dashboard development.

3.5 Filtering Data

Unnecessary or invalid records were filtered out to ensure that only valid accident records were included in the final dataset.

Filtering the data improved data quality and enhanced dashboard performance.

3.6 Creating Age Group

A new calculated column named **Age Group** was created using **Conditional Column** in Power Query.

Drivers were categorized into three age groups:

Driver Age	Age Group
Less than 26	Youth
26–35	Adult
36 Years and Above	Old

3.7 Creating Experience Group

Another calculated column named **Experience Group** was created.

Drivers were classified according to their driving experience.

<u>Driver Experience</u>	<u>Experience Group</u>
---------------------------------	--------------------------------

0–5 Years	Beginner
6–10 Years	Intermediate
More than 10 Years	Experienced

This grouping helps identify whether accident occurrence is related to driver experience.

3.8 Data Quality Improvement

The data preparation process improved the dataset by:

- I. Standardizing data types.
- II. Removing unnecessary records.
- III. Improving consistency.
- IV. Creating analytical categories.
- V. Preparing the data for DAX calculations.
- VI. Enhancing dashboard performance.

3.9 Power Query Workflow

The final Power Query workflow included the following steps:

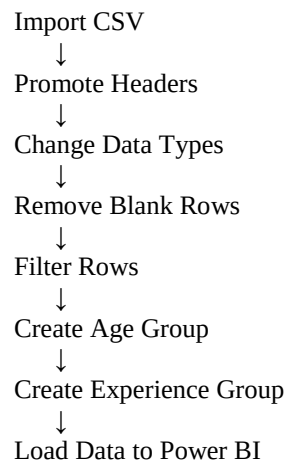
- I. Import CSV Dataset.
- II. Promote Headers.
- III. Change Data Types.
- IV. Remove Blank Rows.
- V. Filter Invalid Records.
- VI. Create Age Group.
- VII. Create Experience Group.
- VIII. Load Data into Power BI Model.

3.10 Benefits of Data Cleaning

Performing data cleaning before analysis provided several benefits:

- I. Improved data accuracy.
- II. Increased dashboard performance.
- III. Better visualization quality.
- IV. Reliable KPI calculations.
- V. Easier data exploration.
- VI. More meaningful analytical insights.

Data Cleaning Workflow



Chapter 4

[DAX Measures]

4.1 Introduction

Data Analysis Expressions (DAX) is the formula language used in Microsoft Power BI to create dynamic calculations, Key Performance Indicators (KPIs), and business metrics.

In this project, DAX measures were created to summarize traffic accident data and provide meaningful insights through interactive dashboards.

The measures automatically update based on slicers and filters, making the dashboard fully interactive.

4.6 Main DAX Measures

Measure Name	DAX Formula	Purpose
Total Accidents	SUM('traffic and accidents'[Accident])	Calculates the total number of accidents recorded in the dataset.
Accident Percentage	DIVIDE([no.of accidents],[No.of rows])	Calculates the percentage of accidents relative to all records.
Average Driver Age	AVERAGE('traffic and accidents'[Driver_Age])	Calculates the average age of drivers involved in accidents.
Average Driver Experience	AVERAGE('traffic and accidents'[Driver_Experience])	Calculates the average driving experience of drivers.
Average Speed Limit	AVERAGE('traffic and accidents'[Speed_Limit])	Calculates the average speed limit across accident records.
Average Number of Vehicles	AVERAGE('traffic and accidents'[Number_of_Vehicles])	Calculates the average number of vehicles involved in accidents.
Maximum Driver Age	MAX('traffic and accidents'[Driver_Age])	Returns the maximum driver age.
Minimum Driver Age	MIN('traffic and accidents'[Driver_Age])	Returns the minimum driver age.
Maximum Speed Limit	MAX('traffic and accidents'[Speed_Limit])	Returns the highest speed limit recorded.

Measure Name	DAX Formula	Purpose
Minimum Speed Limit	MIN('traffic and accidents'[Speed_Limit])	Returns the lowest speed limit recorded.
Alcohol Related Accidents	CALCULATE(COUNTROWS('traffic and accidents'),'traffic and accidents'[Driver_Alcohol]=1)	Counts accidents involving alcohol consumption.
Cases Without Accidents	CALCULATE(COUNTROWS('traffic and accidents'),'traffic and accidents'[Accident]=0)	Counts records where no accident occurred.

4.3 Purpose of Using DAX

DAX measures were developed to provide meaningful KPIs that summarize the dataset and improve dashboard interactivity.

The measures were used to:

- I. Calculate total accident statistics.
- II. Analyze accident distribution.
- III. Calculate averages.
- IV. Compare weather and road conditions.
- V. Study driver behavior.
- VI. Support dashboard filtering.
- VII. Display KPIs dynamically.

4.4 Key Performance Indicators (KPIs)

The following KPIs were displayed on the Executive Dashboard:

- I. Total Accidents
- II. Accident Percentage
- III. Average Driver Age
- IV. Average Driver Experience
- V. Average Speed Limit
- VI. Average Number of Vehicles
- VII. Alcohol Related Accidents

These KPIs provide users with an overall understanding of traffic accident statistics.

Chapter 5
Dashboard Design and Visualization

5.1 Dashboard Overview

The dashboard was designed using Microsoft Power BI to provide an interactive analysis of traffic accident data. The report consists of four dashboard pages, each focusing on a different aspect of traffic accident analysis. Interactive slicers were added to allow users to filter data and explore insights dynamically.

5.2 Dashboard 1 – Executive Overview

Purpose:

This dashboard provides a high-level summary of the traffic accident dataset using Key Performance Indicators (KPIs) and visual charts.

KPIs Displayed:

- I. Total Accidents
- II. Accident Percentage
- III. Alcohol Related Accidents
- IV. Average Driver Age

Visualizations Used:

- I. Card
- II. Clustered Bar Chart
- III. Donut Chart
- IV. Column Chart
- V. Slicers

Business Value:

This dashboard enables users to quickly understand accident statistics and compare accident distribution by road type, weather conditions, and accident severity.



5.3 Dashboard 2 – Driver Analysis

Purpose:

This dashboard focuses on analyzing driver-related factors that may influence accident occurrence.

Visualizations Used:

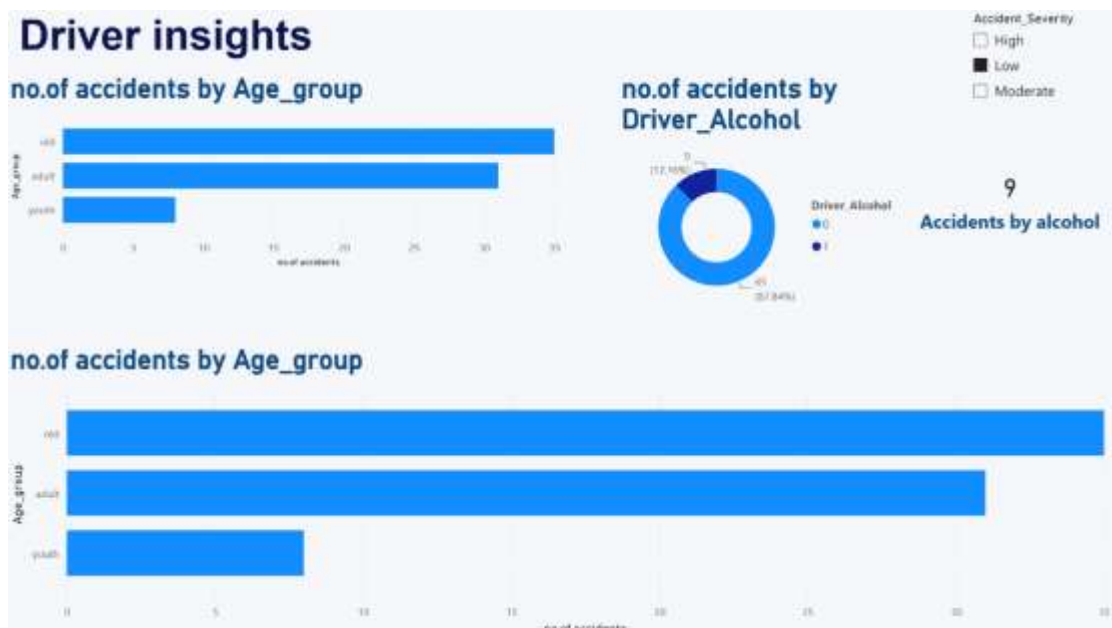
- I. Bar Chart
- II. Donut Chart
- III. KPI Card
- IV. Slicers

Analysis Includes:

- I. Accident distribution by Age Group.
- II. Alcohol-related accidents.
- III. Driver characteristics.

Business Value:

Helps identify the driver groups most frequently involved in accidents and supports awareness and safety initiatives.



5.4 Dashboard 3 – Road & Vehicle Analysis

Purpose:

Analyzes the relationship between road conditions, vehicle types, and traffic density.

Visualizations Used:

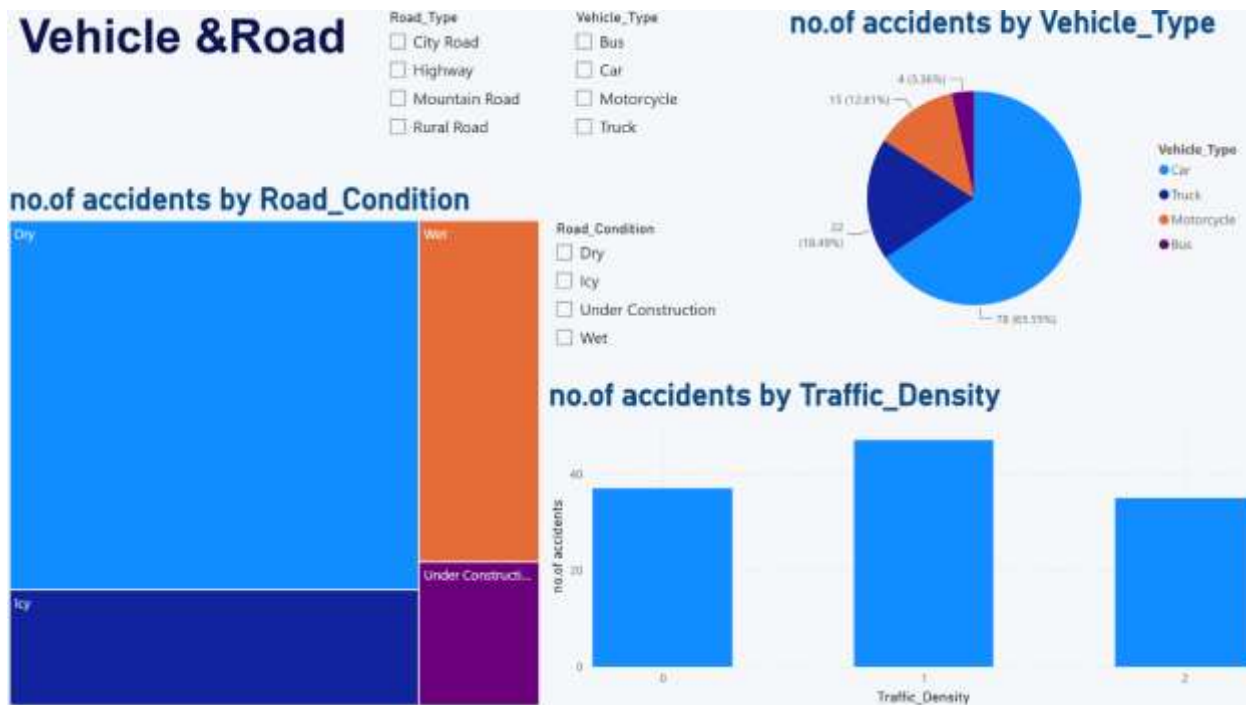
- I. Treemap
- II. Pie Chart
- III. Column Chart
- IV. Slicers

Analysis Includes:

- I. Road Condition.
- II. Vehicle Type.
- III. Traffic Density.

Business Value:

Provides insights into how environmental and road factors contribute to accidents.



5.5 Dashboard 4 – Environment & Time Analysis

Purpose:

Analyzes accidents according to environmental and time-related factors.

Visualizations Used:

- I. Bar Chart
- II. Donut Chart
- III. Line/Column Chart

Analysis Includes:

- I. Time of Day.
- II. Road Light Condition.
- III. Speed Limit.

Business Value:

Helps understand how lighting, time, and speed influence accident occurrence.



5.6 Interactive Features

The dashboard includes several interactive features to improve user experience:

- Slicers for filtering data.
- Cross-filtering between visualizations.
- Dynamic KPI updates.
- Interactive report pages.

5.7 Dashboard Design Principles

The dashboard was designed following Business Intelligence best practices:

- Simple and clean layout.
- Consistent color theme.
- Clear chart titles.
- Appropriate chart selection for each analysis.
- Easy navigation between report pages.
- Interactive filtering capabilities.

Chapter 6

[Key Insights and Findings]

6.1 Overview

After analyzing the traffic accident dataset using Microsoft Power BI, several valuable insights were identified regarding the relationship between accidents and different environmental, road, and driver-related factors.

The interactive dashboard enabled users to explore accident patterns from multiple perspectives and supported data-driven analysis.

6.2 Key Findings

1. Road Type Analysis

The analysis showed that **Highways recorded the highest number of accidents** in this dataset, followed by **City Roads**, while **Mountain Roads** recorded the lowest number of accidents.

This suggests that roads with higher traffic volumes and higher vehicle speeds are associated with more recorded accidents.

2. Weather Analysis

The dashboard indicates that **Clear weather conditions** recorded the largest number of accidents, followed by **Rainy** weather.

This may be because most driving occurs during clear weather rather than because clear weather is inherently more dangerous.

3. Accident Severity

Most accidents were classified as **Low Severity**, while **High Severity** accidents represented the smallest percentage of the dataset.

This indicates that the majority of recorded accidents resulted in limited damage or injuries.

4. Driver Analysis

The analysis revealed that the **Old** age group recorded the highest number of accidents, followed by the **Adult** group, while the **Youth** group recorded the lowest number.

5. Alcohol Consumption

The dashboard shows that only a small proportion of accidents involved drivers who had consumed alcohol, while most recorded accidents involved drivers without alcohol consumption.

6. Vehicle Type

Cars represented the largest proportion of vehicles involved in accidents, followed by trucks and motorcycles.

This may reflect the higher number of cars on the road compared to other vehicle types.

7. Road Condition

Most accidents occurred on **Dry roads**, followed by **Wet roads**, while **Under Construction** roads recorded the fewest accidents.

8. Traffic Density

The dashboard indicates that accidents increase under **Medium Traffic Density**, while lower accident counts were observed under Low and High Traffic Density in this dataset.

9. Road Light Condition

Most recorded accidents occurred under **Artificial Light**, followed by **Daylight**, while **No Light** represented the smallest share.

Chapter 7
[Recommendations]

7.1 Introduction

Based on the dashboard analysis, several recommendations can be made to improve road safety and reduce the likelihood of traffic accidents. These recommendations are derived from the insights obtained through the Power BI dashboard.

7.2 Recommendations

1. Improve Highway Safety

Since highways recorded the highest number of accidents in this dataset, it is recommended to:

- Increase traffic monitoring.
 - Install additional speed cameras.
 - Enhance road safety signs.
 - Strengthen traffic law enforcement.
-

2. Raise Driver Awareness

Awareness campaigns should target drivers by promoting:

- Safe driving practices.
 - Defensive driving techniques.
 - Speed limit compliance.
 - Responsible driving behavior.
-

3. Reduce Speed-Related Risks

As speed is an important factor in accident analysis, it is recommended to:

- Monitor speeding violations.
 - Review speed limits in high-risk areas.
 - Encourage compliance through enforcement.
-

4. Improve Road Conditions

Road maintenance should be performed regularly to improve driving safety, especially on roads affected by weather conditions.

5. Enhance Traffic Management

Traffic congestion can influence accident occurrence. Therefore, improving traffic flow through better traffic management systems may help reduce accident frequency.

6. Improve Road Lighting

Roads with poor lighting should be improved by installing additional lighting systems to increase visibility during nighttime.

7. Continue Data Monitoring

Organizations should continuously collect and analyze accident data using Business Intelligence tools such as Power BI to support data-driven decision-making.

7.3 Expected Benefits

Implementing these recommendations may help:

- I. Improve road safety.
- II. Reduce accident frequency.
- III. Increase driver awareness.
- IV. Support government decision-making.
- V. Improve transportation planning.

Chapter 8
[Conclusion]

8.1 Conclusion

This project presented an interactive Power BI dashboard for analyzing traffic accident data. The dashboard transformed raw accident records into meaningful visual insights through the use of Power Query, DAX measures, and interactive visualizations.

The analysis explored the relationship between accidents and several factors, including weather conditions, road type, driver age, driver experience, vehicle type, traffic density, road conditions, and lighting conditions.

The dashboard enables users to monitor accident statistics, compare different factors, and gain a better understanding of accident patterns through interactive reporting.

Overall, this project demonstrates how Business Intelligence tools can support data-driven decision-making and simplify the analysis of complex datasets.

8.2 Future Work

The dashboard can be enhanced in future versions by:

- I. Integrating real-time accident data.
- II. Developing predictive models using Machine Learning.
- III. Creating geographic maps to visualize accident locations.
- IV. Expanding the dataset with additional years of data.
- V. Publishing the dashboard on Power BI Service for online access and collaboration.

References:

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- II. Microsoft. *Power BI Documentation*.
- III. Microsoft. *Power Query Documentation*.
- IV. Microsoft. *Data Analysis Expressions (DAX) Documentation*.